



ASTROFIX

CELESTIAL NAVIGATION · 100% OFFLINE

User manual

Complete guide to every feature of the app: sextant sight reduction, nautical almanac, compass deviation check and calculation methods, step by step.



Edition in **English** · RDRP Studio · astrofix.app

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Introduction

What ASTROFIX is

ASTROFIX is a celestial navigation app for Android built for a single purpose: to do, on the device itself and without any internet connection, all the desk work that traditionally requires paper, calculator and tables — the **Nautical Almanac**, the **sight reduction tables** and the onboard **calculator**.

ASTROFIX **does not replace the sextant**: you are still the one measuring the altitude of the body above the horizon. What the app does for you is everything that follows that measurement: correcting it, calculating the position of the body at the exact instant of the sight, solving the spherical position triangle, obtaining the intercept, combining several lines of position by least squares and giving you a fix — with the same mathematical rigor a navigator would apply by hand, but in seconds and without calculation errors.

100% offline. All calculation — body positions, magnetic model, time corrections, altitude reduction — runs on the phone itself. No server is contacted, no mobile data or Wi-Fi is needed, and nothing you enter is sent anywhere. It works the same in the middle of the ocean, in airplane mode, as it does in your living room.

The app covers the Sun, the Moon, the four navigational planets (Venus, Mars, Jupiter and Saturn) and the 58 classic navigational stars, including Polaris. Besides sight reduction, it includes a compass deviation checker, a log of saved observations, a Nautical Almanac page exportable to PDF, and a camera- and sensor-assisted star finder.

Teaching philosophy

ASTROFIX was designed both for real use on board and for learning. That's why any fix you calculate can be exported as a **full PDF report** that shows, step by step, exactly the same formulas and intermediate values the app used: the altitude correction, the position of the body as it would be looked up in the almanac, the position triangle and the intercept calculation.

RDRP Studio's recommendation is simple: repeat the calculations by hand, with a calculator or with paper sight reduction tables, and compare your result with the app's before trusting it blindly. It is, by far, the best way to internalize classical celestial navigation — and to have an app that, besides calculating, helps you understand why the result is what it is.

Nautical safety recommendation. Thorough as it is, ASTROFIX is a calculation aid. Always carry a paper Nautical Almanac and printed sight reduction tables as backup for the sextant, in case the device fails, runs out of battery, or is damaged by water or impact.

How to use this manual

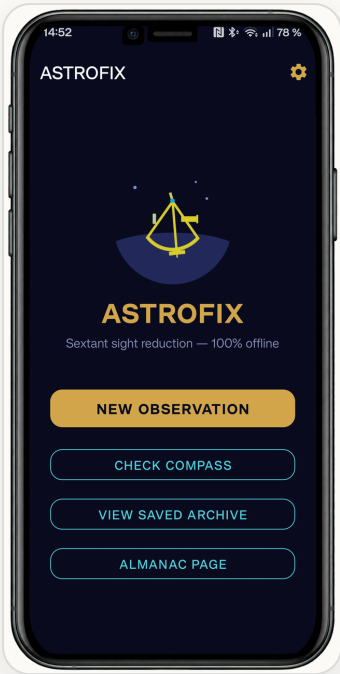
Chapters 2 through 12 walk through the app screen by screen, in the same order you will encounter them when using it: from the home screen, through the general settings, to the complete flow of an observation (setup → measurement capture → summary → result), and the three additional features accessible from Home: check compass, observation log and almanac page.

Chapter 13, *Calculation methods*, brings together in one place all the mathematical and astronomical formulas ASTROFIX uses internally — useful both as a technical reference and for checking your own manual calculations. Chapter 14 is a complete nautical glossary of every technical term that appears in the app, and the final appendices list the full sets of available bodies and constellations.

Getting started

Home screen

Opening ASTROFIX takes you straight to the home screen. From here you can access the app's four main functions, and the gear icon in the top-right corner, which leads to **Settings** (see chapter 3).



Button	What it does
New observation	Starts the wizard for reducing a full celestial observation (chapters 4 to 8).
Check compass	Calculates the compass needle deviation on a given heading by taking a bearing on a body of known position (chapter 9).
View saved log	Opens the log of previously calculated and saved observations (chapter 10).
Almanac page	Generates a PDF with the Nautical Almanac page for the body, date and position you choose (chapter 11).

ASTROFIX home screen.

Step-by-step navigation

The full observation (New observation) is organized as a multi-step wizard — you will see **Step X of 4** at the top of every screen in the process. The navigation buttons are always the same:

- **Back** — returns to the previous step without losing the data entered.
- **Cancel** — abandons the observation in progress and returns to Home.
- **Next / Continue** — validates the current step's data and moves forward.

The wizard's four steps are: **1. Setup** (starting data), **2. Capture measurements** (one or several), **3. Summary** (review before calculating) and **4. Result** (calculated position, with its full breakdown). Each is explained in detail in chapters 5 to 8.

Offline, always. You can close the app in the middle of an observation at sea and resume it later: the data from each step is kept as long as you don't press Cancel. The only data lost between sessions is the observation in progress if you don't save or export it from the Result step.

Settings

Reached via the gear icon in the top-right corner of the home screen. Here you set up, once, the values the app will reuse in every subsequent observation.

Language and default values

ASTROFIX is available in six languages: Spanish, English, German, Italian, French and Portuguese. By default it follows the phone's system language; you can set a different one from this menu at any time.

The **default values** are the starting point automatically filled in when starting a new observation (always editable in the Setup step):

- **Index error** — your sextant's own error, which is subtracted (or added) from every altitude measured with it.
- **Default height of eye (m)** — your usual observation height above sea level, used to calculate the dip of the horizon.

Compass deviation table

If your vessel has a known compass deviation, enter it here as a table of 8 headings (N, NE, E, SE, S, SW, W, NW), with the sign E (+) or W (–) according to your own deviation card or compensation curve. The app automatically interpolates the value for any intermediate heading not listed in the table.

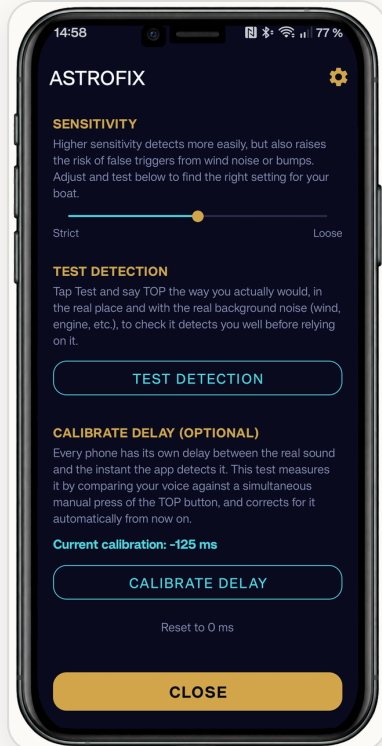
This table is used to calculate the vessel's **true course** (compass + deviation + magnetic declination) when correcting the line of position for the vessel's motion between measurements (see chapter 5). You can also fill it in directly from the **Check compass** function (chapter 9), which calculates the actual deviation by observing a celestial body.

Magnetic model (WMM)

Magnetic declination — the difference between true North and magnetic North at a given place and date — is calculated with NOAA/NGA's official **WMM (World Magnetic Model)**, built into the app. Each version of the model is valid for 5 years; this screen shows which one is currently active and until what date.

When NOAA publishes a newer version, you can **import the official WMM.COF file** directly here, without waiting for an app update. It is downloaded from the official page: ncei.noaa.gov/products/world-magnetic-model. An additional button lets you **restore the built-in model** (WMM2025) at any time.

Leeway from wind — coefficient K



Settings section with the leeway coefficient K, the camera star-finder correction, and access to voice time capture.

Camera star finder — correction

If, when using the camera star finder (see 6.3), the reticle always drifts the same way on your particular phone, compare the diagnostic panel's values against the body's real position and adjust the difference here: **camera azimuth correction (°)** and **camera altitude correction (°)**. This is a factory misalignment between the orientation sensor and the camera's actual optical axis, common on many phone models, and this correction compensates for it permanently.

Voice time capture

This same Settings screen gives access to the voice time capture configuration ("say TOP to fix the time without letting go of the sextant"), an experimental feature explained in detail in chapter 12.

Built-in help and glossary

Settings also includes two quick-reference shortcuts, available at all times: **Help — how to use the app** (a browsable summary of every screen) and the **Nautical glossary** (with the definition of every technical term that appears in the app). The full content of both is included in this manual: the built-in help, spread across the chapters for each screen, and the glossary, in full, in chapter 14.

There is no general physical formula that calculates leeway from wind without first calibrating the boat: every hull, rig and draft reacts differently. The **coefficient K** is only a reasonable starting point — beginning at 5.0 — for the formula:

ESTIMATED LEEWAY

$$\text{leeway} \approx K \cdot \text{wind} \cdot \sin(\text{relative angle}) / \text{speed}$$

Adjust K by comparing, after an observation made in wind, the effective course the app calculated with the actual course measured by GPS: if the estimated leeway is too small, increase K; if it overshoots, lower it.

New observation — overview

"New observation" is the heart of ASTROFIX: the wizard that turns one or more sextant measurements into a fix. It is organized into four sequential steps:

1	Setup	Sextant data, dead-reckoning position (by hand or GPS), vessel motion, wind and current. Filled in once, at the start.
2	Capture measurements	Repeated once per measurement: choose the body, the limb, the altitude read on the sextant and the exact time. You can take as many measurements as you like.
3	Summary	Review of all measurements taken before calculating — any one can be tapped to edit it.
4	Result	The calculated position, the line-of-position chart, the full step-by-step breakdown of each measurement, and the options to save or export to PDF.

How many measurements are needed? There is no fixed number. With a single measurement a position cannot be determined — you only get the line of position relative to the dead-reckoning position — **at least 2** measurements are needed for a fix, and **3 or more**, ideally with clearly different azimuths, for a better-determined and more reliable position.

Chapters 5 to 8 cover each of these four steps in detail.

Setup

The first step of every observation gathers the starting data: the state of your sextant, where you are (dead-reckoning position) and, if the vessel is moving during the measurement session, its course, speed, and the effect of wind and current.

Sextant data

- **Index error** — preloaded from Settings, editable in each observation.
- **Height of eye above sea (m)** — also preloaded from Settings; determines the dip of the horizon.

Dead-reckoning position



Enter your dead-reckoning latitude and longitude with the selector wheels, or tap **Autofill with GPS** to fill it in directly with the phone's GPS position. The magnetic declination for that position is recalculated automatically with the WMM model as soon as you change the position; if it doesn't match your chart, you can correct it by hand.

Location permission is only used to read that single position — it is not shared with anyone or used for any other purpose; the app remains 100% offline.

GPS autofill and, if the vessel is underway, wind and current to take into account.

Vessel underway between measurements

If you're going to take several measurements while the vessel is underway (instead of stopping for each one), turn on the **Vessel underway between measurements?** switch and enter the **compass course** and **speed** — or autofill both with the GPS button (with the caveat that, at very low speed, GPS course is not reliable and only speed is filled in). With this data, ASTROFIX will calculate a **running fix**: it advances each previous measurement to the instant of the last one, exactly as you would do by hand on the chart.

Wind and current

If wind or current affect the vessel's true course, turn them on with their own switch and enter direction and speed for each:

- **Wind** — direction it blows from (°) and speed (knots). Used to estimate **leeway** with the coefficient K set in Settings (see 3, "Leeway from wind").
- **Current** — direction it flows toward (°) and speed (knots). Applied as **drift**, exactly (vectorially), on top of the course already corrected for leeway.

The result of this correction chain is the **effective course and speed** — the vessel's actual motion over the ground — used to calculate the running fix. Section 13.6 explains the full calculation with its formulas.

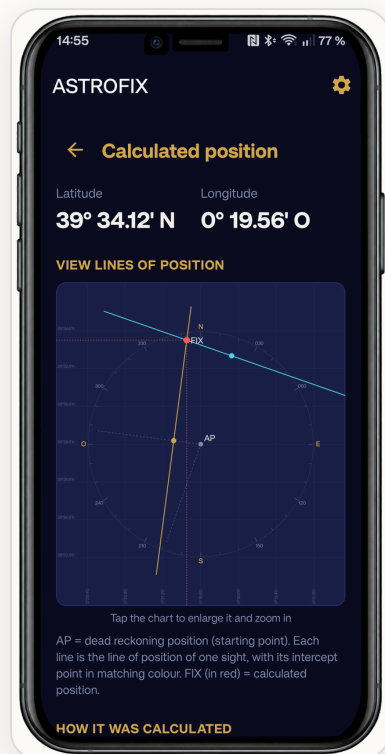
Dead-reckoning date/time

Updates itself when you autofill with GPS. If the dead-reckoning position is from a while ago (for example, the last one fixed a few hours back), edit it by hand so the running-fix calculation correctly advances the position to the instant of the first measurement.

Capturing measurements

This step repeats once for each measurement you take. The screen shows "Measurement X of Y" (or "Take your measurements" if there isn't yet a fixed number) and you can always add another sample or delete the last one.

Choosing the body

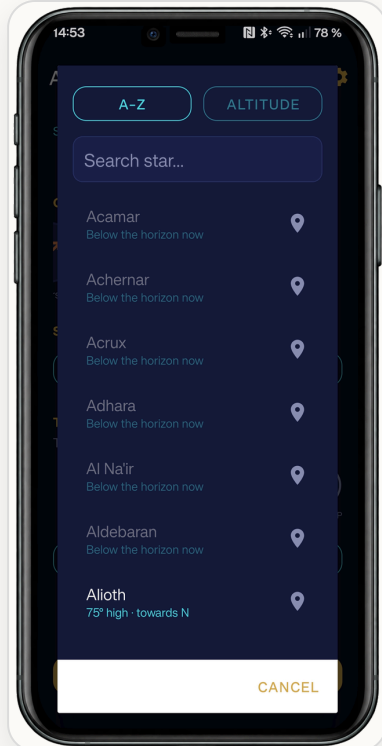


ASTROFIX covers the **Sun**, the **Moon**, the navigational planets **Venus**, **Mars**, **Jupiter** and **Saturn**, and the 58 classic **navigational stars** (full list in appendix 15.1), including Polaris. Bodies that would not be above the horizon right now, given the dead-reckoning position and time entered, appear dimmed in the selection.

To identify a planet with the naked eye, without binoculars: **Venus** is the brightest body in the night sky after the Moon, with a steady light (it doesn't twinkle) and is only visible near dawn or dusk; **Mars** has a clearly noticeable orange or reddish color; **Jupiter** is the third brightest, after the Moon and Venus; **Saturn** shines with a pale yellowish tone, somewhat less intense than Jupiter. None of the four twinkles, unlike stars.

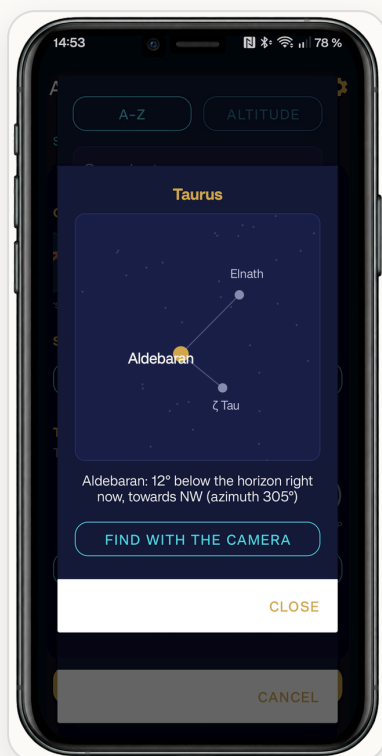
Selecting the observed body, the limb, and capturing the time with the TOP button.

Star catalogue: A–Z and by altitude



Catalogue of the 58 navigational stars, with search and A–Z or current-altitude sorting.

Locating it in the sky

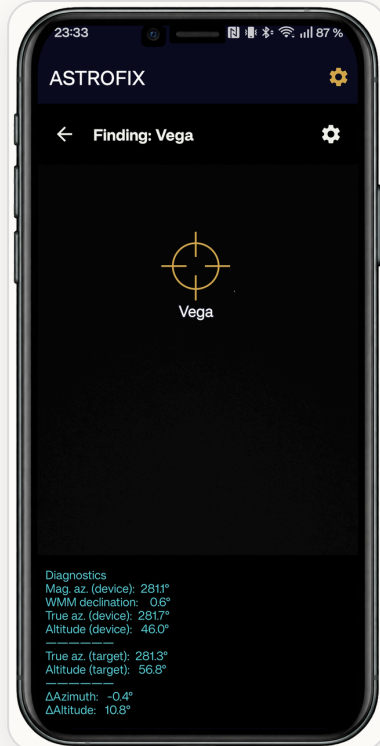


Constellation card: Taurus, with Aldebaran marked, current altitude and azimuth.

Camera star finder

Choosing "Star" opens the full catalogue of the 58 navigational stars. You can sort them **alphabetically (A–Z)** or **by current altitude** above the horizon (highest first, which makes it easier to pick the best candidate at that moment), and search by name directly with the text field. Each row shows its approximate altitude and direction right now, or whether it is below the horizon.

If you don't recognize the star by sight, the **Locate in the sky** button opens a card with a schematic drawing of its constellation, marking the chosen star, and help text: its altitude above the horizon and which direction to look (or whether it is below the horizon right now). The full list of included constellations is in appendix 15.2.



Searching for Vega — the reticle moves in real time with the phone's orientation, guiding you toward the body.

SCREEN VIEW: THREE MODES

Tapping the gear icon on this screen opens a menu with three options:

- **Full screen** — only the camera and the reticle, with no panel overlaid.
- **Diagnostics** — overlays a panel with the raw sensor values: the device's magnetic and true azimuth, the device's altitude, the target body's actual azimuth and altitude, and the difference (Δ) between the two in azimuth and altitude. Useful for checking whether the finder is well calibrated on your phone (see 3, "Camera star finder").
- **Calibrate reticle** — replaces the diagnostics panel with two **on-screen sliders**, one for azimuth correction and one for altitude (range -15.0° to $+15.0^\circ$, in steps of 0.1°). Each slider always starts from the value already saved in Settings, and the reticle reacts in real time as you move them: it's a visual, immediate way to find the right adjustment, instead of trying values blindly in Settings. The value is saved automatically at all times — even if you leave the screen without switching modes again — so no separate "Save" button is needed.

Just a visual aid. The camera finder, in any of its three modes, helps you orient toward the body. Measuring the altitude is still always done with the sextant.

From any body's card, the **Find with camera** button opens a live view combining the camera image with the phone's orientation sensors: a reticle moves across the screen, pointing exactly toward where the chosen body is at each instant. It's an aid to **find** it by eye or with the sextant — it does not replace the sextant for the measurement itself.

If the body is outside the camera's field of view, the reticle is replaced by an arrow at the edge of the screen, with text showing how many degrees to turn (azimuth) and raise or lower (altitude) to reach it.

Capturing measurements (cont.)

Observed limb

When measuring the Sun or Moon with the sextant you touch its edge, not its center — indicate whether you used the **upper limb** or the **lower limb**. The app automatically applies the corresponding semidiameter correction (+SD with lower limb, −SD with upper limb). It does not apply to stars and planets, since they are measured as a point.

Sextant altitude (Hs)

Enter the value read directly on the sextant, in degrees and minutes — the raw reading, with no correction applied yet; ASTROFIX takes care of the whole correction chain (see chapter 13.1).

Time of the measurement (TOP)

The exact instant of the measurement matters just as much as the altitude itself, because the body's position depends on it. There are three ways to set it: tapping the TOP button, entering the time by hand, or turning on voice capture:

- **TOP button** — press the large button at the exact instant you bring the body down onto the horizon with the sextant.
- **Enter measurement manually** — if you prefer to note the time with a separate stopwatch, you can type it in by hand (year, month, day, hour, minute, second).
- **Voice capture (experimental)** — say "TOP" out loud and the app detects the instant without you letting go of the sextant or taking your hands off to press anything. Explained in detail in chapter 12.

Next to the captured UTC time, the **approximate local time** is also shown (nautical time zone: UTC ± 1h per 15° of longitude) — useful on long passages crossing several time zones, where UTC alone doesn't tell you much at a glance.

Several measurements: same body or different bodies

You can take more than one measurement in a row, either of the **same body** (for example, several altitudes of the Sun within a few minutes) or of **different bodies** (for example, Sun, Venus and a star in the same session) to obtain more lines of position and a more precise fix. In both cases, use **Add sample** / **Remove sample**; the "Measurements taken" counter keeps track. To switch bodies, simply return to the **Choose the body** section and select the next one before taking the new measurement. When something goes wrong, **Retake capture** lets you re-measure that same sample without losing the previous ones.

Vessel motion during capture

If the course or speed changes **during** the measurement session — for example, you stop the vessel to take the next measurement more precisely, or you change course between one star and another — record it in the **Vessel motion** section on this same screen (date/time from which it applies, new course and speed; speed 0 if the vessel has stopped). Wind and current stay the same as in Setup

unless you change them here. ASTROFIX will take every recorded leg into account when calculating the final running fix (see 13.6).

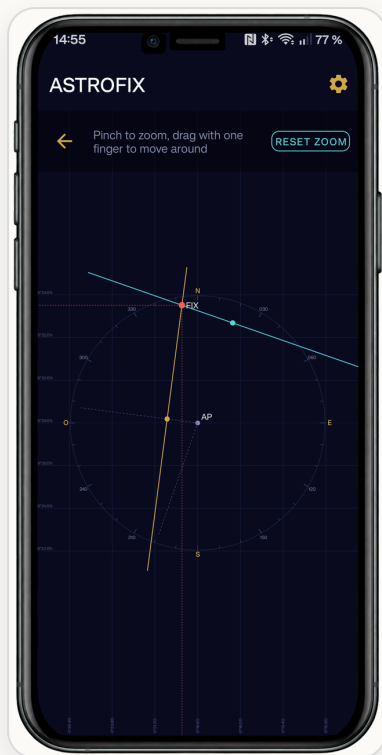
Summary of measurements

Before calculating, ASTROFIX shows a summary of every measurement taken: body, time and altitude of each. **Tap any measurement to edit it** — in case you spot an error while reviewing — or go back to add another. When you're satisfied, **Calculate position** moves on to the final step.

Last check before calculating. This is the best moment to make sure no time capture is missing and that the entered altitudes make sense (for example, that the wrong limb wasn't entered) — an error here propagates straight through to the final result.

Result

Calculated position

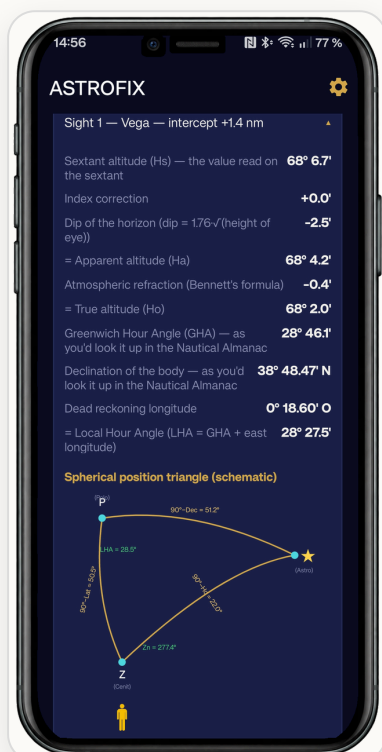


Calculated position, with latitude, longitude and the line-of-position chart.

The Result screen shows the **calculated latitude and longitude**, combining every measurement from the previous step. If only one measurement was taken (or several with nearly identical azimuths, which don't add different geometry), a real fix cannot be determined: the app warns you and shows the uncorrected dead-reckoning position instead.

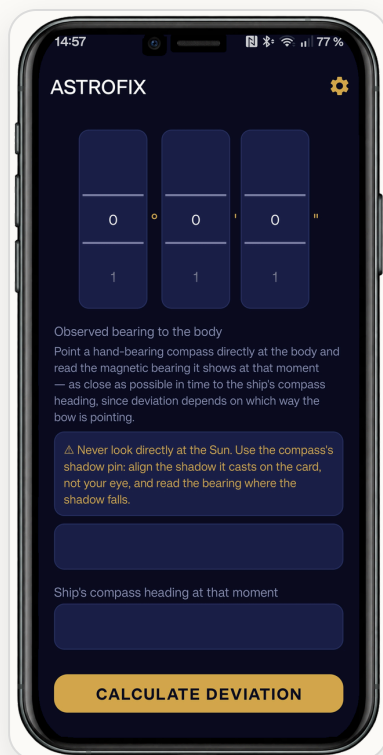
From here you can **compare with the phone's GPS** (difference in nautical miles) and **compare with the starting dead-reckoning position** (also in nautical miles) — two quick ways to gauge the quality of the fix obtained.

Line-of-position chart



Enlarged chart: pinch to zoom, drag with one finger to pan.

The chart shows the dead-reckoning position (AP), each line of position (one per measurement, with its intercept point in the matching color) and the final calculated position (FIX, in red), in the same style you would plot by hand on a chart. Tap it to expand to full screen, where you can zoom with two fingers and pan by dragging with one — useful for examining closely how well (or poorly) the lines cross.



Breakdown of a measurement: altitude correction, body position and spherical position triangle.

Tapping **How it was calculated**, and then any measurement in the list, expands its full breakdown: the altitude correction ($H_s \rightarrow H_o$), the body's position as it would be looked up in the nautical almanac (GHA, declination, LHA), the spherical position triangle drawn schematically, and the calculation of H_c , Z_n and the intercept — with the exact formulas visible at each step. If there were legs of vessel motion, the correction for leeway and current applied is also shown here.

All the formulas shown on this screen are explained in more detail in chapter 13.

Save and export to PDF

From the Result screen you can:

- **Save to the log** — to look it up later from "View saved log" (chapter 10).
- **Export as PDF** — with two versions to choose from:
 - **Summary PDF (logbook)** — a compact single page with the dead-reckoning position, the setup used, the table of measurements and the final result. Meant for the ship's logbook.
 - **Full PDF (educational)** — the extended report, with the complete breakdown of each measurement (the same formulas and steps shown on screen) and the final chart of the lines of position. Meant for practice: print it and try to repeat the calculations by hand before looking at the already-solved values.

Sight 1 — Vega

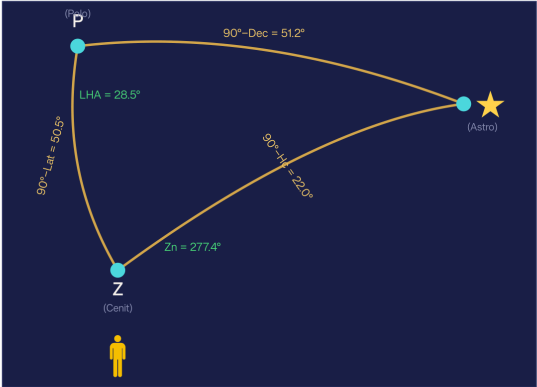
20:30:00 UTC — 2026-09-21 (20:30 (UTC+0))
Altitude correction (Hs → Ho)

Sextant altitude (Hs) — the value read on the sextant	68°6.7'
Index correction	+0.0'
Dip of the horizon (dip = 1.76·√(height of eye))	-2.5'
= Apparent altitude (Ha)	68°4.2'
Atmospheric refraction (Bennett's formula)	-0.4'
= True altitude (Ho)	68°2.0'

Body's position (as in the Nautical Almanac)

Greenwich Hour Angle (GHA) — as you'd look it up in the Nautical Almanac	28°46.1'
Declination of the body — as you'd look it up in the Nautical Almanac	38°48.5' N
Dead reckoning longitude	0°18.6' O
= Local Hour Angle (LHA = GHA + east longitude)	28°27.5'

Position triangle and Hc/Zn calculation



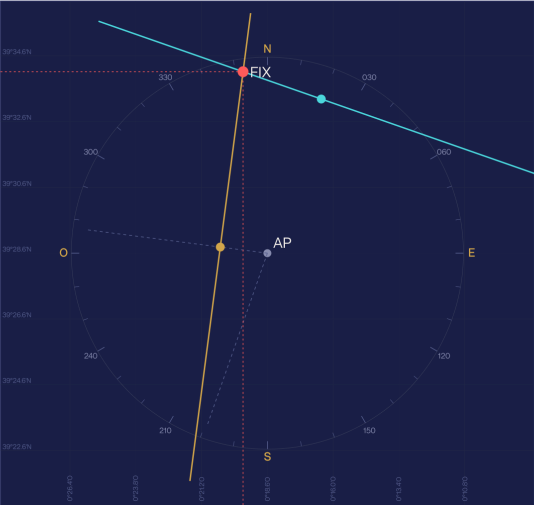
ASTROFIX — RDRP Studio — page 2

Actual page from the full PDF report, with the position triangle worked out step by step.

How the lines are combined

Each sight gives a line of position (perpendicular to its azimuth, at the intercept distance). With 2 or more lines, the final position is the point that best crosses all of them at once — calculated by least squares, the same principle as crossing the lines by hand on a chart.

Graphical plot of the lines of position



Result

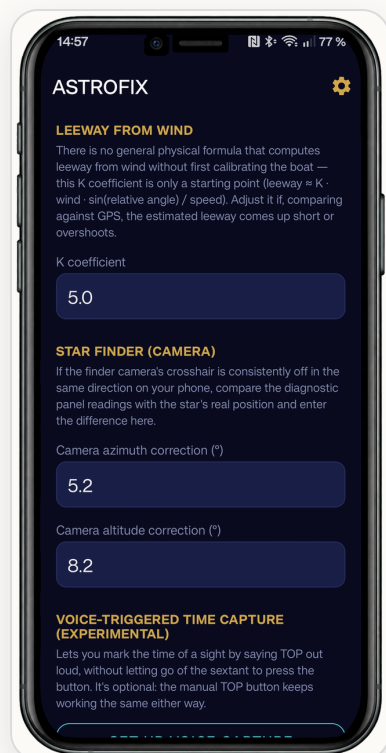
Latitude: 39°34.1' N
Longitude: 0°19.6' O

Fix correctly determined.

Document generated by ASTROFIX (RDRP Studio) for teaching purposes — compare your manual calculations against the ones shown here. Always keep a printed Nautical Almanac and reduction tables as a backup to the sextant.
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Actual page from the full PDF report, with the chart of the lines of position and the final fix.

Check compass



Compass check: bearing observed on the body and vessel's compass course.

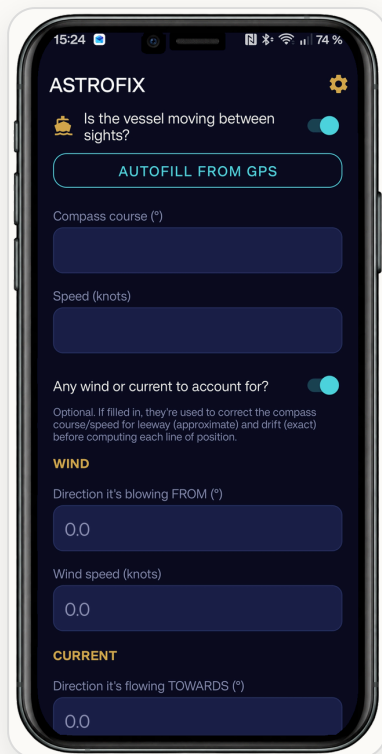
This function, accessible from Home, calculates the **compass needle deviation** on a specific heading, with no extra instruments needed: just take a bearing on a body of known position (Sun, Moon or a planet) with a bearing compass.

1. Choose the body to take a bearing on.
2. Point a bearing compass directly at the body and read the **magnetic bearing** it shows at that moment — as simultaneously as possible with the vessel's compass course, because deviation depends on where the bow is pointing.
3. Enter that **observed bearing** and the vessel's **compass course** at that instant.
4. Tap **Calculate deviation**.

⚠ **Never look directly at the Sun.** Use the compass's shadow pin: line up the shadow it casts on the card — not your line of sight — and read the heading where that shadow falls.

The app calculates the **true bearing** to the body (from its actual position and yours), subtracts the **magnetic declination** (WMM) to obtain the expected magnetic bearing, and compares it with the magnetic bearing you actually observed: the difference is the **needle deviation** on that heading. The result can be **saved directly to the deviation table** in Settings, on the nearest of the 8 available headings.

Observation log

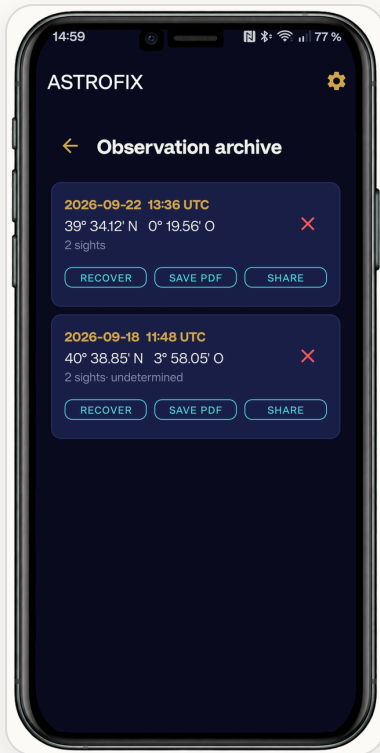


Log of saved observations, with date, position and number of measurements of each.

Every observation you save from the Result screen (chapter 8) is kept here, sorted by date, with its calculated position and the number of measurements used. On each entry:

- **Retrieve** — loads it again and shows its result, with the Back arrow available to review its original measurements and setup.
- **Save PDF** — exports that observation (summary or full, as in chapter 8).
- **Share** — sends the PDF through any app installed on the phone (mail, messaging, etc.). The app remembers where it was last saved, to overwrite without asking again.
- **Delete** — removes the observation from the log (with confirmation; cannot be undone).

Almanac page



Selecting bodies and correction tables to include in the almanac PDF.

Body(ies) to include

You can choose one or several of: Sun, Moon, Venus, Mars, Jupiter, Saturn. Each generates its own hourly table of **GHA** (plus Aries) and **Declination**, with the interpolation columns **v** and **d**, for all 24 hours of the chosen day.

Generates a PDF with a **paper-style Nautical Almanac page** — calculated with exactly the same formulas the rest of the app uses — for the body, date and position you choose. Useful as a printed backup, or to check your own manual calculations without needing the annual Almanac volume.

Hour	GHA Aries	GHA Mars	Dec Mars	v	d
00h	3 2"	2437"	2°35 9'	+09'	-03'
01h	18 2"	2587"	2°35 7'	+09'	-03'
02h	33 2"	273 8"	2°35 4'	+09'	-03'
03h	48 3"	288 8"	2°35 2'	+09'	-03'
04h	63 3"	303 8"	2°34 9'	+09'	-03'
05h	78 4"	318 8"	2°34 6'	+09'	-03'
06h	93 4"	333 8"	2°34 4'	+09'	-03'
07h	108 4"	348 8"	2°34 1'	+09'	-03'
08h	123 5"	3 8"	2°33 9'	+09'	-03'
09h	138 5"	18 9"	2°33 6'	+09'	-03'
10h	153 6"	33 9"	2°33 4'	+09'	-03'
11h	168 6"	48 9"	2°33 1'	+09'	-03'
12h	183 6"	63 9"	2°32 9'	+09'	-03'
13h	198 7"	78 9"	2°32 6'	+09'	-03'
14h	213 7"	93 9"	2°32 4'	+09'	-03'
15h	228 8"	108 9"	2°32 1'	+09'	-03'
16h	243 8"	124 0'	2°31 9'	+09'	-03'
17h	258 9"	139 0'	2°31 6'	+09'	-03'
18h	273 9"	154 0'	2°31 4'	+09'	-03'
19h	288 9"	169 0'	2°31 1'	+09'	-03'
20h	304 0'	184 0'	2°30 9'	+09'	-03'
21h	319 0'	199 0'	2°30 6'	+09'	-03'
22h	334 1'	214 1'	2°30 3'	+09'	-03'
23h	349 1'	229 1'	2°30 1'	+09'	-03'

Actual page of the almanac PDF generated by ASTROFIX, with Mars's hourly table.

Correction tables to include

Optionally, the PDF can also include the classic correction tables from the printed Nautical Almanac, already solved: altitude correction for the Sun (upper/lower limb), for stars/Jupiter/Saturn, additional parallax correction for Venus and Mars, Moon correction (based on horizontal parallax HP), dip of the horizon by height of eye, arc-to-time conversion, and the summary table of GHA increments per minute. The last is included only as a reference for using the paper almanac — internally, ASTROFIX always calculates GHA exact to the second, with no manual interpolation needed.

NAVIGATIONAL STARS

Turning on **Navigational stars (58, SHA/Dec)** adds a page with the Sidereal Hour Angle and Declination of the 58 navigational stars, just like the star page of the printed Nautical Almanac.

SUNRISE, SUNSET AND TWILIGHT

Adds the time of sunrise and sunset for the Sun and Moon, the start and end of civil and nautical twilight, and the equation of time — all for the given position and date. At very high latitudes it indicates when the body does not rise (polar night) or does not set (midnight sun) that day.

Voice time capture (TOP)

Experimental, optional feature. The manual TOP button on the Capture screen always keeps working the same way, whether this feature is turned on or not.

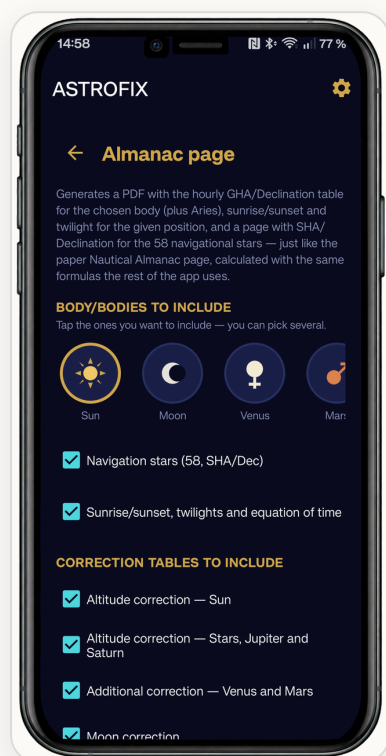
Lets you fix the time of a measurement by saying "TOP" out loud, without letting go of the sextant to press the button — useful at the critical instant you bring the body down onto the horizon. It compares the sound picked up by the microphone against recorded templates of your own voice saying "TOP": it is **not** general speech recognition, uses no network or external services, and runs entirely on the phone itself.

Configured from Settings → **Configure voice capture**.

Samples of your voice

Record the word "TOP" several times, in the voice you would actually use when taking a measurement (even somewhat raised, as if there were wind) — the more samples, the better detection adapts to how you say it. At least 3 samples are needed to enable listening; the maximum is 6 (remove one if you want to record a different one).

Sensitivity



The sensitivity slider ranges from **Strict** to **Permissive**. Higher sensitivity detects "TOP" more easily, but also increases the risk of false triggers from wind noise, bumps or other sounds. Adjust and test with the function below until you find the right setting for your boat and its usual conditions.

Sensitivity, detection test and delay calibration.

Testing detection

Press **Test detection** and say "TOP" the way you really would, on location and with real background noise (wind, engine, etc.), to confirm it detects you well before relying on the feature during an actual

observation. The result shows whether it was detected and with what match percentage against your templates.

Calibrate delay (optional)

Every phone has its own delay between the actual sound and the instant the app detects it. This test measures it: press the TOP button exactly as you say "TOP", three times in a row — as you would with the sextant in hand — and the app calculates the average difference and applies it automatically from then on, correcting the time captured by voice. It can be reset to 0 ms at any time.

Calculation methods

This chapter brings together, with their exact formulas, all the mathematical and astronomical calculation ASTROFIX performs internally for every observation — the same one shown, step by step, on the "How it was calculated" screen and in the full (educational) PDF report. It's recommended to follow along with a calculator in hand and reproduce the calculation of a real measurement from the app, comparing each intermediate value.

13.1 Altitude correction ($H_s \rightarrow H_o$)

The altitude read directly on the sextant (**H_s** , sextant altitude) is not the body's actual altitude above the true horizon: it must be corrected in several steps to obtain the **true altitude (H_o)**, which is the one actually used in calculating the position.

STEP 1 – INDEX CORRECTION

$H_a' = H_s + \text{index correction}$

STEP 2 – DIP OF THE HORIZON

$\text{dip} = 1.76 \cdot \sqrt{(\text{height of eye in meters})}$ [minutes of arc, negative]

= APPARENT ALTITUDE

$H_a = H_a' + \text{dip}$

STEP 3 – ATMOSPHERIC REFRACTION (BENNETT'S FORMULA)

$R = 1 / \tan(H_a + 7.31/(H_a + 4.4))$ [minutes of arc, always negative]

STEP 4 – SEMIDIAMETER (SUN AND MOON ONLY)

$\pm SD \rightarrow +SD$ with lower limb, $-SD$ with upper limb

STEP 5 – PARALLAX IN ALTITUDE (SUN, MOON AND PLANETS)

$\text{parallax} = HP \cdot \cos(H_a)$ [always positive]

= TRUE ALTITUDE

$H_o = H_a + R + SD + \text{parallax}$

Refraction is always negative (the body appears higher than it actually is) and depends on the apparent altitude: it is much greater near the horizon than near the zenith. Semidiameter only applies to the Sun and Moon, whose disk is large enough that you measure the limb rather than the center; for stars, Jupiter and Saturn the parallax correction is negligible (less than 0.05') and is not applied — it is, however, relevant for the Sun, the Moon and, to a lesser extent, Venus and Mars, whose distance from Earth varies noticeably with the date.

13.2 Position of the body at the instant of the measurement

With the exact UTC time captured (TOP), ASTROFIX calculates internally, for any body and any instant, exactly the same data that would be looked up in the paper Nautical Almanac:

GREENWICH HOUR ANGLE

GHA — the body's hour angle relative to the Greenwich meridian, at the exact instant of the measurement

DECLINATION

Dec – the celestial equivalent of latitude: the body's angular distance from the celestial equator

= LOCAL HOUR ANGLE

$LHA = GHA + \text{longitude (E positive, W negative)}$

For the Sun, the Moon and the planets, these positions are derived from the astronomical ephemerides built into the app (equivalent to those published in the annual Nautical Almanac). For the 58 navigational stars, it starts from their catalogued Right Ascension and Declination (referred to the J2000 equinox, with the corresponding precession applied) and from the GHA of Aries at the instant of the measurement:

$GHA(\text{star}) = GHA(\text{Aries}) + SHA(\text{star})$

Calculation methods (cont.)

13.3 Spherical position triangle

Sight 1 — Vega
2025-03-01T00:00:00Z - 2025-03-01T00:00:00Z (UTC-08)
Altitude correction (Hs → Ho)

Sextant altitude (Hs) — the value read on the sextant	68°6.7'
Index correction	+0.0'
Dip of the horizon (dip = 1.76 × √(height of eye))	-2.5'
→ Apparent altitude (Ha)	68°4.2'
Atmospheric refraction (Barnett's formula)	-0.4'
→ True altitude (Hc)	68°2.0'

Body's position (as in the Nautical Almanac)

Greenwich Hour Angle (GHA) — as you'd look it up in the Nautical Almanac	28°46.1'
Declination of the body — as you'd look it up in the Nautical Almanac	38°48.5' N
Dead-reckoning longitude	0°18.6' O
→ Local Hour Angle (LHA = GHA + east longitude)	28°27.5'

Position triangle and Hc/Zn calculation

The **position triangle** is the spherical triangle formed by three vertices: the elevated **Pole** (P), the observer's **Zenith** (Z) and the observed **Body**. Its three sides and angles relate the dead-reckoning latitude, the body's declination and the local hour angle to the body's altitude and azimuth — it is the basis of the entire line-of-position calculation, both by hand with sight reduction tables and in ASTROFIX.

The triangle's three sides are 90°–Lat (Pole–Zenith), 90°–Dec (Pole–Body) and 90°–Hc (Zenith–Body); the angle at the Pole is the LHA, and the angle at the Zenith is the azimuth Zn.

The spherical position triangle Pole–Zenith–Body, as drawn by ASTROFIX in the breakdown of each measurement (real example, Vega).

COMPUTED ALTITUDE (Hc)

$$\sin(Hc) = \sin(Lat) \cdot \sin(Dec) + \cos(Lat) \cdot \cos(Dec) \cdot \cos(LHA)$$

AZIMUTH (ZN)

$$Zn = \text{atan2}(-\sin(LHA), \cos(Lat) \cdot \tan(Dec) - \sin(Lat) \cdot \cos(LHA))$$

Hc is the altitude the body *should* have if the observer were exactly at the dead-reckoning position (Lat, Lon) at the instant of the measurement — it is calculated by spherical trigonometry from that position and the body's position (GHA/Dec) obtained in the previous section. **Zn** is the direction, measured from true North clockwise (0°–360°), toward where the body was at that instant.

13.4 Intercept and line of position

INTERCEPT

$$\text{Intercept} = (Ho - Hc) \times 60 \quad [\text{nautical miles}]$$

The intercept is the difference between the true altitude actually measured (Ho) and the altitude computed for the dead-reckoning position (Hc), converted to nautical miles (1 minute of arc = 1 nautical mile). If Ho > Hc, the ship is closer to the body than the dead-reckoning position suggests: the intercept is **toward the body**. If Ho < Hc, it is **away from the body**.

Plotting that number of miles from the dead-reckoning position, in the direction of the azimuth Zn (toward or away, according to the sign), gives a point through which the **line of position** passes — perpendicular to the azimuth at that point. The actual ship is somewhere on that line; a single measurement is not enough to know where.

How the lines are combined

Each sight gives a line of position (perpendicular to its azimuth, at the intercept distance). With 2 or more lines, the final position is the point that best crosses all of them at once — calculated by least squares, the same principle as crossing the lines by hand on a chart.

Graphical plot of the lines of position



Result

Latitude: 39°34.1' N
Longitude: 0°19.6' O

Fix correctly determined.

Document generated by ASTROFIX (RORP Studio) for teaching purposes — compare your manual calculations against the ones shown here. Always keep a printed Nautical Almanac and reduction tables as a backup to the software.
2025-08-10 10:10:10 UTC — 14217

Actual chart of two lines of position crossing at the FIX — example generated by ASTROFIX.

With two or more measurements you get two or more lines of position, each with its own azimuth and intercept. The final position (**FIX**) is the point that best crosses all the lines at once.

With exactly two non-parallel lines, the FIX is simply their intersection point. With three or more, manual practice looks for the "triangle of error" and estimates the center by eye; ASTROFIX, instead, solves the overdetermined system by **least squares**: the point that minimizes the sum of the squares of the perpendicular distances to all the lines simultaneously — the same principle as crossing the lines by hand on a chart, but solved exactly and objectively. If the geometry of the lines is poorly conditioned (azimuths nearly parallel to each other), the system is left underdetermined and the app flags it explicitly instead of giving a misleading position.

Calculation methods (cont.)

13.6 Running fix: leeway, drift and effective course

When the vessel moves between measurements (chapter 5), ASTROFIX advances each earlier line of position to the instant of the last one by means of a **running fix**, calculated with the vessel's real effective course and speed over the ground:

STEP 1 – TRUE COURSE

$R_v = \text{Compass course} + \text{Deviation (table, interpolated)} + \text{Magnetic declination (WMM)}$

STEP 2 – LEEWAY FROM WIND (APPROXIMATE)

$\text{Leeway} \approx K \cdot V(\text{wind}) \cdot \sin(\text{relative wind angle}) / V(\text{vessel})$

= COURSE OVER SURFACE

$R_s = R_v + \text{Leeway}$

STEP 3 – DRIFT FROM CURRENT (EXACT, VECTORIAL)

The vector (course over surface, own speed) is added vectorially to the current's vector (direction, speed)

= EFFECTIVE COURSE AND SPEED

R_e, V_e – the vessel's real motion over the ground

The correction for **deviation** and **declination** is exact (tables and the WMM model). The correction for **leeway** is necessarily approximate, because there is no universal physical formula that derives it from the wind without first calibrating the specific boat — hence the adjustable coefficient K in Settings (see 3 and 13.6). The correction for **current**, on the other hand, is exact: a simple vector composition of velocities. If there are several distinct course/speed legs during the session (recorded in the Capture step), this same chain is applied to each leg separately before advancing the line.

13.7 Needle deviation — compass check

The "Check compass" function (chapter 9) reverses part of this chain to *measure* the actual deviation, instead of assuming it from a table:

TRUE BEARING TO THE BODY

calculated from its GHA/Dec and the dead-reckoning position, the same as the azimuth Z_n (13.3)

EXPECTED MAGNETIC BEARING

= True bearing to the body – Magnetic declination (WMM)

NEEDLE DEVIATION

= Observed magnetic bearing – Expected magnetic bearing

13.8 Magnetic declination — WMM model

Magnetic declination — the difference between true North and magnetic North at a given point and date — is obtained from NOAA/NGA's official **World Magnetic Model (WMM)**, a mathematical model of the Earth's magnetic field expressed as a series of spherical harmonics of degree and order 12, valid for 5-year cycles. ASTROFIX evaluates it entirely on the device from the dead-reckoning position (latitude, longitude) and the date of the observation, with no external lookup — and lets you import a newer version of the coefficients file (WMM.COF) as soon as NOAA publishes it (see chapter 3).

13.9 Nautical time-zone time

Alongside every UTC time, ASTROFIX also shows the approximate local time according to the traditional **nautical time zone**: a 15°-of-longitude zone per hour, with no dependence on any database of administrative time zones (which, obviously, doesn't exist on the open sea).

```
Nautical zone (hours) = round(Longitude / 15°) - sign + to the East, - to the West
```

Nautical glossary

The same glossary built into the app (Settings → Nautical glossary), with the definition of every technical term that appears in ASTROFIX, grouped by topic.

POSITION TRIANGLE AND COORDINATES OF THE BODY

Azimuth (Zn) — Angle, measured from true North clockwise (0° – 360°), that marks the direction to the body at the moment of the sight.

Altitude (H) — The angle between the body and the horizon. In the app it appears as Hs (sextant altitude, the one you read on the sextant), Ho (true altitude, after corrections) and Hc (computed altitude, what your dead-reckoning position 'should' give).

Zenith — The point directly above the observer, on the local vertical; one of the three vertices of the position triangle.

Elevated pole — The Earth's pole (North or South) that lies above the observer's horizon; the second vertex of the triangle.

Position triangle — The spherical triangle Pole–Zenith–Body that relates latitude, declination and hour angle to the body's altitude and azimuth; it is the basis of the entire line-of-position calculation.

Declination (of the body) — The celestial equivalent of latitude: the body's angular distance from the celestial equator.

Local hour angle (LHA) — How far the body has shifted, in longitude, from the observer's meridian.

Right ascension — The celestial equivalent of longitude, used together with declination to locate the body in the sky.

LINE OF POSITION AND FIX

Line of position — The line of position resulting from a single altitude measurement; the ship is somewhere on that line.

Intercept — The distance (in miles) between your dead-reckoning position and the line of position, and whether you must move 'toward' or 'away from' the body to reach it.

Dead-reckoning position (DR) — The position calculated from the last known point, the course and the speed, without correction from any observation.

Fix — The actual position, obtained by crossing two or more lines of position.

Nautical glossary (cont.)

SEXTANT AND MEASUREMENT CORRECTIONS

Index error — Each sextant's own error, which must be subtracted (or added) from every altitude measured with it.

Height of eye — The observer's height above sea level; the dip-of-the-horizon correction depends on it.

Dip — The correction for height of eye: the higher the observer, the more the visible horizon 'drops'.

Refraction — The bending the body's light undergoes as it passes through the atmosphere, which makes it appear higher than it actually is.

Parallax — The correction for measuring from the Earth's surface rather than from its center (relevant only for the Sun, Moon and planets).

Semidiameter / Limb — When measuring the Sun or Moon with the sextant, you touch its edge (upper or lower limb), not its center; semidiameter corrects for that difference.

COURSE AND CORRECTION FOR WIND/CURRENT

Bearing — The direction, measured with a compass, toward an object (a body, the coast, another ship).

True course — The course relative to true (geographic) North, before applying compass deviation.

Needle deviation — The onboard compass's own error, different on each heading, caused by the ship's own magnetism.

Magnetic declination — The difference between true North and magnetic North at a given place and date (calculated here with the WMM model).

Leeway — The sideways drift the wind causes on the ship's course.

Drift (from current) — The additional deviation, in course and speed, caused by the current.

Course over surface / Effective course — The course already corrected for leeway (over surface) and then also for current (effective): the one the ship actually follows over the ground.

TIME

UTC — Universal time, the reference for all astronomical calculations.

Nautical zone time — The approximate local time based on longitude (UTC $\pm$ a 15° zone), with no dependence on any time-zone database.

Glossary of calculation formulas

The same glossary built into the app (Help → Calculation formulas), with the data required and the standard celestial-navigation formula for each calculation ASTROFIX performs, grouped by topic.

SEXTANT ALTITUDE CORRECTION

Calculating the dip of the horizon:

he — observer's height of eye above sea level (m)

FORMULA:

$\text{Dip} = 1.76 \times \sqrt{\text{he}}$ (arcminutes; he in metres)

Calculating atmospheric refraction:

Ha — apparent altitude of the body, after index error and dip (°)

FORMULA:

$R = 0.97127 / \tan(\text{Ha}) - 0.00137 / \tan^3(\text{Ha})$ (arcminutes)

Refraction is always subtracted from the altitude.

Calculating the semi-diameter (Sun and Moon):

- HP — Moon's equatorial horizontal parallax, from the ephemeris (')
- R — Earth–Sun distance in astronomical units, from the ephemeris

FORMULA:

$\text{SD Moon} = 0.2724 \times \text{HP}$

$\text{SD Sun} \approx 16.0' / R$

Only applies to the Sun and the Moon; added for lower limb, subtracted for upper limb.

Calculating parallax in altitude:

- HP — body's equatorial horizontal parallax, from the ephemeris (')
- Ha — apparent altitude of the body (°)

FORMULA:

$\text{Parallax} = \text{HP} \times \cos(\text{Ha})$

Added to the altitude. Only significant for the Sun, the Moon, Venus and Mars; for stars, Jupiter and Saturn it is negligible ($< 0.05'$) and is not applied.

Calculating the true altitude (Ho):

- Hs — sextant altitude, read on the instrument (°)
- IC — sextant index correction (°)
- Dip — dip of the horizon (°)
- R — atmospheric refraction (°)
- SD — semi-diameter, if applicable (°)

- P — parallax in altitude, if applicable (°)

FORMULA:

$$Ha' = Hs + IC$$

$$Ha = Ha' - Dip$$

$$Ho = Ha - R \pm SD + P \text{ (+SD for lower limb, -SD for upper limb)}$$

POSITION OF THE BODY

Calculating the GHA of Aries:

Exact UTC time of the sight (TOP)

FORMULA:

Derived from Greenwich Apparent Sidereal Time (GAST) for the exact UTC instant of the sight. Not a one-line formula: it is computed with a full astronomical development, the same as in the annual Nautical Almanac.

Calculating the GHA and declination of the Sun, Moon and planets:

Exact UTC time of the sight (TOP); date of the observation

FORMULA:

Derived from the body's astronomical ephemeris for the exact instant of the sight, using the same celestial-mechanics formulas as the annual Nautical Almanac — not summarised in a short formula.

Calculating the GHA of a star:

- GHA Aries — Greenwich hour angle of Aries at the instant of the sight (°)
- SHA — sidereal hour angle of the star, from the catalogue (°)

FORMULA:

$$GHA(star) = GHA(Aries) + SHA(star)$$

Calculating the Local Hour Angle (LHA):

- GHA — Greenwich hour angle of the body (°)
- Lon — longitude of the DR position (° East +, West -)

FORMULA:

$$LHA = GHA + Lon \text{ (East positive, West negative; normalised between } 0^\circ \text{ and } 360^\circ)$$

Glossary of calculation formulas (cont.)

POSITION TRIANGLE AND LINE OF POSITION

Calculating the computed altitude (Hc):

- Lat — latitude of the DR position (°)
- Dec — declination of the body (°)
- LHA — local hour angle (°)

FORMULA:

$$\sin(Hc) = \sin(Lat) \cdot \sin(Dec) + \cos(Lat) \cdot \cos(Dec) \cdot \cos(LHA)$$

Calculating the azimuth (Zn):

- Lat — latitude of the DR position (°)
- Dec — declination of the body (°)
- Hc — computed altitude (°)
- LHA — local hour angle (°)

FORMULA:

$$\cos(Z) = (\sin(Dec) - \sin(Lat) \cdot \sin(Hc)) / (\cos(Lat) \cdot \cos(Hc))$$

If LHA > 180°, Zn = Z; if LHA < 180°, Zn = 360° - Z

Calculating the intercept:

- Ho — true (observed) altitude, measured and corrected (°)
- Hc — computed altitude for the DR position (°)

FORMULA:

$$\text{Intercept} = (Ho - Hc) \times 60 \text{ (nautical miles)}$$

If Ho > Hc, toward the body; if Ho < Hc, away from the body.

Calculating the fix with several lines of position — least squares:

- Zn i — azimuth of each line of position (°)
- Intercept i — intercept of each line of position (nautical miles)

FORMULA:

Each line i contributes one equation:

$$\cos(Zn\ i) \cdot \Delta lat + \sin(Zn\ i) \cdot \cos(Lat) \cdot \Delta lon = \text{Intercept } i$$

With 3 or more lines, the system is overdetermined; ASTROFIX solves it by least squares, finding the point (Δlat , Δlon) that minimises the sum of squared distances to all the lines.

COURSE, LEEWAY/DRIFT AND COMPASS

Calculating the true course (Rv):

- Compass course — course read on the ship's compass (°)

- Deviation — from the deviation table, interpolated (° East +, West –)
- Magnetic declination — from the WMM model for the position and date (° East +, West –)

FORMULA:

$R_v = \text{Compass course} + \text{Deviation} + \text{Magnetic declination}$

Calculating leeway from wind (coefficient K):

- K — the ship's leeway coefficient, adjustable in Settings (default 5.0)
- Wind — apparent wind speed
- Relative angle — angle between the ship's heading and the wind direction
- Speed — ship's speed

FORMULA:

$\text{Leeway} \approx K \cdot \text{Wind} \cdot \sin(\text{Relative angle}) / \text{Speed}$

This is an approximation: there is no universal physical formula without first calibrating the specific vessel.

Calculating set and drift from current:

- R_v — true course of the ship (°)
- V_e — ship's speed through the water
- R_c — direction the current is setting toward (°)
- V_c — current speed

FORMULA:

$V_x = V_e \cdot \sin(R_v) + V_c \cdot \sin(R_c)$

$V_y = V_e \cdot \cos(R_v) + V_c \cdot \cos(R_c)$

$\text{Course made good} = \text{atan2}(V_x, V_y)$

$\text{Speed made good} = \sqrt{V_x^2 + V_y^2}$

Calculating the running fix advance:

- Speed made good — resulting from course/current/leeway (knots)
- Course made good — resulting from course/current/leeway (°)
- Δt — time elapsed between the earlier sight and the latest one (hours)

FORMULA:

$\text{Distance run} = \text{Speed made good} \times \Delta t \text{ (nautical miles)}$

The earlier line of position is advanced by that distance in the direction of the course made good, to the instant of the latest sight.

Calculating compass deviation (compass check):

- Z_n — true azimuth of the observed body, computed (°)
- Magnetic declination — from the WMM model (° East +, West –)
- Compass bearing — bearing to the body read on the ship's compass (°)

FORMULA:

$\text{Magnetic bearing} = Z_n - \text{Magnetic declination}$

$$\text{Deviation} = \text{Magnetic bearing} - \text{Compass bearing}$$

Glossary of calculation formulas (cont.)

MAGNETISM AND TIME

Calculating magnetic declination:

- Lat, Lon — DR position (°)
- Date — date of the observation
- WMM model coefficients (WMM.COF file, NOAA/NGA)

FORMULA:

Evaluated with the WMM (World Magnetic Model), a spherical-harmonic series of degree and order 12 – not summarised in a short formula. Input: position and date. Output: magnetic declination at that point.

Calculating the nautical zone time:

Lon — longitude of the DR position (° East +, West –)

FORMULA:

Zone (hours) = round(Lon / 15°) (+ sign East, – West)

Appendices

15.1 Available bodies

Sun, Moon, Venus, Mars, Jupiter, Saturn, and the 58 classic navigational stars:

- | | |
|--------------|-------------------|
| · Acamar | · Fomalhaut |
| · Achernar | · Gacrux |
| · Acrux | · Gienah |
| · Adhara | · Hadar |
| · Aldebaran | · Hamal |
| · Alioth | · Kaus Australis |
| · Alkaid | · Kochab |
| · Al Na'ir | · Markab |
| · Alnilam | · Menkar |
| · Alphard | · Menkent |
| · Alphecca | · Miaplacidus |
| · Alpheratz | · Mirfak |
| · Altair | · Nunki |
| · Ankaa | · Peacock |
| · Antares | · Polaris |
| · Arcturus | · Pollux |
| · Atria | · Procyon |
| · Avior | · Rasalhague |
| · Bellatrix | · Regulus |
| · Betelgeuse | · Rigel |
| · Canopus | · Rigil Kentaurus |
| · Capella | · Sabik |
| · Deneb | · Schedar |
| · Denebola | · Shaula |
| · Diphda | · Sirius |
| · Dubhe | · Spica |
| · Elnath | · Suhail |
| · Eltanin | · Vega |
| · Enif | · Zubenelgenubi |

Appendices (cont.)

15.2 Included constellations

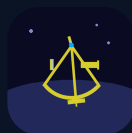
Constellation cards available in "Locate in the sky" (chapter 6) for the navigational stars:

- Andromeda
- Phoenix
- Cassiopeia
- Whale (Cetus)
- Eridanus
- Aries
- Perseus
- Taurus
- Orion
- Charioteer (Auriga)
- Keel (Carina)
- Great Dog (Canis Major)
- Little Dog (Canis Minor)
- Gemini
- Sails (Vela)
- Hydra
- Leo
- Great Bear (Ursa Major)
- Crow (Corvus)
- Southern Cross (Crux)
- Virgo
- Centaurus
- Herdsman (Bootes)
- Libra
- Little Bear (Ursa Minor)
- Northern Crown (Corona Borealis)
- Scorpius
- Southern Triangle (Triangulum Australe)
- Ophiuchus
- Dragon (Draco)
- Sagittarius
- Lyra
- Eagle (Aquila)
- Peacock (Pavo)
- Swan (Cygnus)
- Pegasus
- Crane (Grus)
- Southern Fish (Piscis Austrinus)

15.3 PDF report format

ASTROFIX generates four types of PDF document, all with the same footer ("ASTROFIX — RDRP Studio — page X") and calculated with exactly the same formulas as the rest of the app:

Document	Contents	Where it's generated
Summary PDF (logbook)	Dead-reckoning position, setup, table of measurements and result, on one compact page.	Result / Log
Full PDF (educational)	Full step-by-step breakdown of each measurement, with formulas, position triangle and final chart of lines of position.	Result / Log
Almanac page	Hourly GHA/Dec tables, navigational stars, sunrise/sunset/twilight and classic correction tables.	Almanac page (Home)



RDRP Studio

ASTROFIX is an app by RDRP Studio — independent software development. This manual describes the version of ASTROFIX current as of the publication date shown on the cover; some features marked as experimental (voice time capture) may evolve in future versions.

All observation reduction, the WMM magnetic model and the astronomical ephemerides are calculated entirely on the device — ASTROFIX does not connect to any server or share data with third parties.